

Mahmoud Abdelhamid

AI Engineer

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[Portfolio](#) - [linkedin](#) - [Kaggle](#)

WORK EXPERIENCE

AI Engineer

ThinkTech Company

Jun 2025 - Ongoing

SaudiLendGateAI — AI Financial Statement Analyzer

- Built an asynchronous FastAPI RAG service that analyzes corporate financial documents (PDF/XLSX/Word) for Saudi lending and credit decisions, delivering 14+ specialized analysis types (audited financials, credit scoring, market research) as structured JSON.
- Engineered the ingestion pipeline using Google Document AI OCR, OpenAI embeddings, and Qdrant retrieval, with a thread-safe job queue, webhook callbacks, and Dockerized Azure DevOps CI/CD across dev/prod.

MCP-Borsa — Multi-Agent Brokerage Chat Backend

- Designed a Dapr-based multi-agent backend where a supervisor agent delegates to specialized Knowledge (ChromaDB RAG) and Database (read-only SQL) sub-agents, streaming answers to the frontend over Server-Sent Events (SSE).
- Implemented a custom Model Context Protocol (MCP) client and integrated postgres-mcp for guarded SQL access; durable agent workflows checkpoint to Redis, deployed on Azure AKS via ArgoCD GitOps.

Baseera — AI Company Research Platform

- Built a full-stack platform (Nuxt 4 on Cloudflare Workers edge) that analyzes 1,000+ companies per batch using domain-restricted agentic web search via the OpenAI Responses API and Tavily.
- Architected a pluggable search/LLM provider layer with browser-side concurrent batch processing and Langfuse tracing on Cloudflare D1/R2, eliminating server-side queue cost.

Company Website Refactor: Refactored the company website using AI-assisted rapid-prototyping tools (Replit, Lovable) to ship UI changes quickly. **Tools & Technologies:** Python, FastAPI, TypeScript, Nuxt 4, Dapr, Model Context Protocol (MCP), LangChain, LangGraph, OpenAI GPT-4.1/GPT-5 APIs, OpenAI embeddings, Qdrant, ChromaDB, postgres-mcp, PostgreSQL, Redis, Google Document AI, Tavily, Exa, Langfuse, LangSmith, Pydantic, Docker, Azure Pipelines, Azure AKS, ArgoCD, Cloudflare Workers/D1/R2, Replit, Lovable.

Remote AI & ML Engineer

Upwork, Direct Clients, Udemy, Udacity

July 2022 - Ongoing

SmartClaim / Hakeem AI — Medical Insurance Claim Validation (Backend Lead)

- Owned the backend of an AI system that validates medical-insurance prescriptions: a FastAPI service that extracts structured data from prescription PDFs/images via Google Gemini Vision OCR, then auto-approves, rejects, or flags each claim against clinical and policy rules with bilingual (English/Arabic) output.
- Took it from POC to production by re-architecting into an event-driven RabbitMQ pipeline (OCR + validation workers with dead-letter queues and idempotent retries) backed by async PostgreSQL (SQLAlchemy 2.0 / Alembic) and dual local/S3 storage.
- Built 13+ REST endpoints spanning case intake, OCR, clinical validation, decisioning with override audit trails, and multi-tenant accounts, secured with JWT/RBAC and covered by 80+ unit and integration tests.

Other Freelance & Teaching Work

- Delivered diverse AI & ML projects for global clients across predictive modeling, NLP, computer vision, and generative AI, often leading small teams and working directly with stakeholders.
- Transitioned into teaching, delivering AI courses and workshops on Udemy and Udacity to mentor aspiring engineers.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, C++, R, SQL

Generative AI & Agents: LangChain, LangGraph, Dapr Agents, CrewAI, Phidata, DSPy, Model Context Protocol (MCP), RAG, Multi-Agent Systems, Prompt Engineering, Pydantic, Transformers (BERT, GPT, LLaMA, Mixtral)

LLM Providers & Serving: OpenAI (GPT-4.1/GPT-5), Google Gemini (Vertex AI), Groq, Hugging Face, LLM APIs
Vector DBs & Retrieval: Qdrant, ChromaDB, FAISS, OpenAI Embeddings, Semantic Search
OCR & Document AI: Google Document AI, PyMuPDF, Unstructured, PaddleOCR, Tesseract, YOLO (DocLayNet)
Deep Learning & NLP: TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, spaCy, NLTK
Backend & APIs: FastAPI, Flask, Streamlit, Server-Sent Events (SSE), Async Python, RabbitMQ, SQLAlchemy
Cloud & DevOps: Docker, Kubernetes, Google Cloud, ArgoCD, CI/CD, Linux, Git, GitHub Actions
Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite3
Evaluation & Monitoring: Langfuse, Ragas, LangSmith
Data & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn
Core Concepts: RAG, Multi-Agent Systems, Microservices, Message Queues, Neural Networks, CNN, RNN, LSTM, Computer Vision, NLP, API Development, OOP

PROJECTS

- **OrionIntel — Modular RAG Platform:** A FastAPI RAG platform for analyzing company files, finance reports, and books. Built a layered architecture with LangChain document loaders and chunking, async MongoDB (Motor) persistence, and pluggable LLM (OpenAI/Cohere) and vector-DB (Qdrant) provider factories powering semantic search; Dockerized with pre-commit code-quality tooling.
Tools: Python, FastAPI, LangChain, MongoDB (Motor), Qdrant, OpenAI, Cohere, Docker [GitHub](#)
- **Plant Disease CNN Classifier:** Built a CNN model to detect plant diseases from images with high accuracy. Delivered via an interactive Streamlit app for farmers and researchers.
Tools: Python, TensorFlow, Keras, Streamlit [Try it!](#)
- **Multi-Agent Financial Research Assistant:** Designed a Phidata-based multi-agent system where Groq LLMs coordinate web and finance agents. Delivered Markdown-formatted insights with tables, news context, and market analysis.
Tools: Python, Phidata, Groq LLMs, YFinance, DuckDuckGo [GitHub](#)
- **Graduation Project – MOA Prediction:** Built a full-stack ML platform predicting drug mechanisms of action (Kaggle Lish-MOA). Integrated autoencoder-based deep learning, ensemble models, and a Flask API with a Next.js frontend.
Tools: Python (TensorFlow, XGBoost, scikit-learn), Flask, Next.js, PostgreSQL [GitHub](#)
- **AnimeGPT-LLMOps (Personal Project):** Created a retrieval-augmented anime recommender using ChromaDB and Groq LLMs. Added Streamlit UI, Langfuse observability, Docker/K8s deployment, and Grafana Cloud monitoring.
Tools: Python, Streamlit, LangChain, ChromaDB, Groq, Docker, Kubernetes, Grafana [GitHub](#)

EDUCATION

Menoufia University
B.S. in Computer Science

Egypt – Menoufia
May 2020 - May 2024

GPA: 3.65/4.0 **Achieved** Excellent with Honors **Ranked** 7th in the college

EXTRACURRICULAR ACTIVITIES

- **Kaggle Expert:** Gained expert-level skills by competing in data science challenges, specializing in NLP, ML, Gen AI, Computer Vision, and Deep Learning.
- **Egyptian Collegiate Programming Contest (ECPC):** Participated from 2020 to 2024, then transitioned to coaching in 2024 to mentor aspiring programmers in the competition.
- **Problem-Solving Training:** Solved 1,000+ problems on platforms like Codeforces and Leetcode, with a focus on algorithms and data structures. Regularly competed in 20+ high-level contests, achieving a top 10% placement in several.

PERSONAL INFORMATION

- **Date of Birth:** 23 December 2002 **Military Status:** Exempted **Marital Status:** Single **Nationality:** Egyptian